

CBSE Class 10 Mathematics — Arithmetic Progressions — Solutions

1. 1. The 4th term of an AP is 14 and the 12th term is 70. What is the first term a ?

Answer: (A)

Solution:

1. Use the formula $a_n = a + (n - 1)d$
2. Set up two equations: $a + 3d = 14$ and $a + 11d = 70$
3. Subtract the equations to find $d = 7$
4. Substitute d to find a

Derivation:

$$\begin{aligned}a + 3d &= 14 \\a + 11d &= 70 \\(a + 11d) - (a + 3d) &= 70 - 14 \\8d &= 56 \\\implies d &= 7 \\a + 3(7) &= 14 \\\implies a &= 14 - 21 = -7\end{aligned}$$

Arithmetic Progressions — n th term of an AP

2. 2. If the sum of first n terms of an AP is $S_n = 3n^2 + n$, what is the common difference d ?

Answer: (C)

Solution:

1. Find $S_1 = a_1 = 3(1)^2 + 1 = 4$
2. Find $S_2 = a_1 + a_2 = 3(2)^2 + 2 = 14$
3. Find $a_2 = S_2 - S_1 = 10$
4. Find $d = a_2 - a_1 = 10 - 4 = 6$

Derivation:

$$\begin{aligned}a_1 &= S_1 = 4 \\S_2 &= 3(4) + 2 = 14 \\a_2 &= S_2 - S_1 = 14 - 4 = 10 \\d &= a_2 - a_1 = 10 - 4 = 6\end{aligned}$$

Arithmetic Progressions — Sum of n terms

3. 3. Which term of the AP: 21, 18, 15, ... is -81?

Answer: (B)

Solution:

1. Identify $a = 21$ and $d = -3$
2. Use $a_n = a + (n - 1)d = -81$
3. Solve for n

Derivation:

$$21 + (n - 1)(-3) = -81$$

$$(n - 1)(-3) = -102$$

$$n - 1 = 34$$

$$n = 35$$

Arithmetic Progressions — n th term of an AP

4. **4.** In an AP, if $a = 5$ and $d = 3$, find the sum of the first 10 terms (S_{10})?

Answer: (C)

Solution:

1. Use formula $S_n = \frac{n}{2}[2a + (n - 1)d]$
2. Substitute $n = 10, a = 5, d = 3$

Derivation:

$$S_{10} = \frac{10}{2}[2(5) + (10 - 1)3]$$

$$S_{10} = 5[10 + 27]$$

$$S_{10} = 5 \times 37 = 185$$

Arithmetic Progressions — Sum of n terms

5. **5.** If $k, 2k - 1, 2k + 1$ are three consecutive terms of an AP, find the value of k .

Answer: (A)

Solution:

1. Use property $b - a = c - b$ for three terms a, b, c in AP
2. Solve $(2k - 1) - k = (2k + 1) - (2k - 1)$

Derivation:

$$k - 1 = 2$$

$$k = 3$$

Arithmetic Progressions — Properties of AP

6. **6.** Find the sum of all odd numbers between 0 and 50.

Answer: (B)

Solution:

1. Identify AP: 1, 3, ..., 49
2. Find n using $a_n = a + (n - 1)d$
3. Use $S_n = \frac{n}{2}(a + l)$

Derivation:

$$\begin{aligned}49 &= 1 + (n - 1)2 \\ \implies 48 &= 2(n - 1) \\ \implies n &= 25 \\ S_{25} &= \frac{25}{2}(1 + 49) = \frac{25}{2}(50) = 625\end{aligned}$$

Arithmetic Progressions — Sum of n terms

7. **7.** The 10th term from the end of the AP: 5, 8, 11, ..., 185 is:

Answer: (D)

Solution:

1. Reverse the AP: 185, 182, ..., 5
2. New $a = 185, d = -3, n = 10$
3. Calculate a_{10}

Derivation:

$$\begin{aligned}a_{10} &= 185 + (10 - 1)(-3) \\ a_{10} &= 185 - 27 = 158\end{aligned}$$

Arithmetic Progressions — n th term from the end

8. **8.** If the sum of first n terms is n^2 , then the n th term is:

Answer: (C)

Solution:

1. $a_n = S_n - S_{n-1}$
2. Substitute $S_n = n^2$ and $S_{n-1} = (n - 1)^2$

Derivation:

$$\begin{aligned}a_n &= n^2 - (n - 1)^2 \\ a_n &= n^2 - (n^2 - 2n + 1) = 2n - 1\end{aligned}$$

Arithmetic Progressions — Sum of n terms

9. **9.** How many two-digit numbers are divisible by 3?

Answer: (A)

Solution:

1. First number is 12, last is 99
2. Use $a_n = a + (n - 1)d$ where $d = 3$

Derivation:

$$\begin{aligned}99 &= 12 + (n - 1)3 \\ 87 &= (n - 1)3 \\ 29 &= n - 1 \\ \implies n &= 30\end{aligned}$$

Arithmetic Progressions — Application of n th term

10. 10. What is the common difference of an AP in which $a_{25} - a_{20} = 45$?

Answer: (B)

Solution:

1. Express terms as $a + 24d$ and $a + 19d$
2. Subtract the two expressions

Derivation:

$$(a + 24d) - (a + 19d) = 45$$

$$5d = 45$$

$$\implies d = 9$$

Arithmetic Progressions — n th term of an AP

11. 11. Assertion (A): The common difference of the AP $5, 2, -1, -4, \dots$ is -3 . Reason (R): The common difference d of an AP $a_1, a_2, a_3, \dots$ is given by $d = a_n - a_{n-1}$.

Answer: (A)

Solution:

1. Identify terms $a_1 = 5, a_2 = 2$.
2. Calculate $d = 2 - 5 = -3$.
3. Verify Reason definition of common difference.

Derivation:

$$d = a_2 - a_1 = 2 - 5 = -3$$

Arithmetic Progressions — Common Difference

12. 12. Assertion (A): If the n^{th} term of an Arithmetic Progression is given by $a_n = 3 + 4n$, then the common difference of the AP is 4. Reason (R): The common difference d of an AP is given by the difference between consecutive terms $a_n - a_{n-1}$.

Answer: (A)

Solution:

1. Find the first term a_1 by substituting $n = 1$ in $a_n = 3 + 4n$.
2. Find the second term a_2 by substituting $n = 2$ in $a_n = 3 + 4n$.
3. Calculate the common difference $d = a_2 - a_1$.
4. Observe that the coefficient of n in a linear expression of n represents the common difference.

Derivation:

$$a_1 = 3 + 4(1)$$

$$a_2 = 3 + 4(2)$$

$$d = a_2 - a_1 = 11 - 7$$

Since $d = 4$, Assertion is true

The formula $d = a_n - a_{n-1}$ is the standard definition, so Reason is true and is the correct explanation.

Arithmetic Progressions — general term of an AP

- 13. 13.** Assertion (A): The sum of the first n terms of an arithmetic progression is given by $S_n = \frac{n}{2}[2a + (n - 1)d]$, where a is the first term and d is the common difference. Reason (R): The sum of the first n terms of an AP can be calculated as $S_n = \frac{n}{2}(\text{first term} + \text{last term})$.

Answer: (B)

Solution:

1. The formula $S_n = \frac{n}{2}[2a + (n - 1)d]$ is the standard formula for the sum of n terms of an AP, making Assertion (A) true.
2. The formula $S_n = \frac{n}{2}(a + l)$ where l is the last term (a_n) is also a valid and standard formula for the sum of n terms, making Reason (R) true.
3. While both are true, Reason (R) is a different formulaic representation and does not act as the logical explanation for the derivation of the formula provided in the Assertion.

Derivation:

$$S_n = \frac{n}{2}(a + a_n)$$

$$\implies S_n = \frac{n}{2}(a + a + (n - 1)d) = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic Progressions — sum of n terms of an AP

- 14. 14.** Assertion (A): The sum of the first n terms of an arithmetic progression is given by $S_n = 3n^2 + 5n$. The common difference of this AP is 6. Reason (R): For an AP where $S_n = an^2 + bn$, the common difference is always $2a$.

Answer: (A)

Solution:

1. To find the common difference from the sum of n terms, calculate S_1 and S_2 .
2. $S_1 = a_1 = 3(1)^2 + 5(1) = 8$.
3. $S_2 = a_1 + a_2 = 3(2)^2 + 5(2) = 12 + 10 = 22$.
4. Calculate $a_2 = S_2 - S_1 = 22 - 8 = 14$.
5. Common difference $d = a_2 - a_1 = 14 - 8 = 6$. Assertion is true.
6. For a general sum $S_n = An^2 + Bn$, $a_n = S_n - S_{n-1} = A(n^2 - (n - 1)^2) + B(n - (n - 1)) = A(2n - 1) + B = 2An - A + B$. Since a_n is a linear function of n , the common difference is the coefficient of n , which is $2A$. Reason is true and correctly explains the assertion.

Derivation:

$$S_n = 3n^2 + 5n$$

$$a_n = S_n - S_{n-1}$$

$$a_n = 3n^2 + 5n - [3(n - 1)^2 + 5(n - 1)]$$

$$a_n = 3n^2 + 5n - [3(n^2 - 2n + 1) + 5n - 5]$$

$$a_n = 3n^2 + 5n - 3n^2 + 6n - 3 - 5n + 5$$

$$a_n = 6n + 2$$

$$d = a_n - a_{n-1} = (6n + 2) - (6(n - 1) + 2) = 6$$

Arithmetic Progressions — Sum of first n terms

15. **15.** Assertion (A): The 10th term of the AP: 5, 8, 11, 14, ... is 32. Reason (R): The n th term of an AP with first term a and common difference d is given by $a_n = a + (n - 1)d$.

Answer: (A)

Solution:

1. Identify $a = 5$, $d = 3$, $n = 10$.
2. Calculate $a_{10} = 5 + (10 - 1) \times 3 = 5 + 27 = 32$.
3. The formula in R is the standard n th term formula of an AP.

Derivation:

$$a_{10} = 5 + 9(3) = 5 + 27 = 32. \text{The formula } a_n = a + (n - 1)d \text{ is the standard definition.}$$

Arithmetic Progressions — n th term of an AP

16. **16.** Find the 11th term of an Arithmetic Progression (AP) whose first term is -3 and the common difference is 4.

Answer:

37

Solution:

1. Identify the given parameters: first term $a = -3$ and common difference $d = 4$.
2. Use the formula for the n^{th} term: $a_n = a + (n - 1)d$ and substitute $n = 11$.

Derivation:

$$a_{11} = a + (11 - 1)d = -3 + 10(4) = -3 + 40 = 37$$

Arithmetic Progressions — N th term of an AP

17. **17.** Which term of the AP 21, 18, 15, ... is -81 ?

Answer:

35th term

Solution:

1. Identify $a = 21$ and $d = 18 - 21 = -3$.
2. Set $a_n = -81$ and solve for n using $a_n = a + (n - 1)d$.

Derivation:

$$\begin{aligned} -81 &= 21 + (n - 1)(-3) \\ \implies -102 &= (n - 1)(-3) \\ \implies n - 1 &= 34 \\ \implies n &= 35 \end{aligned}$$

Arithmetic Progressions — N th term of an AP

18. 18. Find the sum of the first 10 multiples of 5.

Answer:

$$275$$

Solution:

1. The AP is 5, 10, 15, ... where $a = 5, d = 5, n = 10$.
2. Use the sum formula $S_n = \frac{n}{2}[2a + (n - 1)d]$.

Derivation:

$$S_{10} = \frac{10}{2}[2(5) + (10 - 1)5] = 5[10 + 45] = 5[55] = 275$$

Arithmetic Progressions — Sum of first n terms

19. 19. If the n^{th} term of an AP is given by $a_n = 3 + 2n$, find its common difference.

Answer:

$$2$$

Solution:

1. Find the first term a_1 by substituting $n = 1$.
2. Find the second term a_2 by substituting $n = 2$, then calculate $d = a_2 - a_1$.

Derivation:

$$a_1 = 3 + 2(1) = 5; \quad a_2 = 3 + 2(2) = 7; \quad d = a_2 - a_1 = 7 - 5 = 2$$

Arithmetic Progressions — General term of an AP

20. 20. Find the value of k for which the terms $k - 1, k + 3, 3k - 1$ are in Arithmetic Progression.

Answer:

$$k = 5$$

Solution:

1. Use the property that if a, b, c are in AP, then $2b = a + c$.
2. Substitute the terms and solve the linear equation for k .

Derivation:

$$\begin{aligned} 2(k + 3) &= (k - 1) + (3k - 1) \\ \implies 2k + 6 &= 4k - 2 \\ \implies 8 &= 2k \\ \implies k &= 5 \end{aligned}$$

Arithmetic Progressions — Properties of AP

- 21. 21.** Find the sum of all two-digit numbers which, when divided by 4, leave 1 as a remainder.

Answer:

$$1225$$

Solution:

1. Identify the sequence: The smallest two-digit number leaving remainder 1 when divided by 4 is 13 ($4 \times 3 + 1$), and the largest is 97 ($4 \times 24 + 1$).
2. The sequence is 13, 17, 21, ..., 97, which is an AP with $a=13$, $d=4$, and $l=97$.
3. Find the number of terms n using the formula $l = a + (n-1)d$.
4. Calculate the sum using $S_n = n/2 \times (a + l)$.

Derivation:

$$97 = 13 + (n - 1)4$$

$$\implies 84 = (n - 1)4$$

$$\implies n - 1 = 21$$

$$\implies n = 22. S_{22} = \frac{22}{2}(13 + 97) = 11 \times 110 = 1225.$$

Arithmetic Progressions — Sum of n terms

- 22. 22.** If the sum of the first n terms of an AP is $S_n = 3n^2 + 5n$, find the 16th term of the AP.

Answer:

$$98$$

Solution:

1. Use the relation between the n th term and the sum of terms: $a_n = S_n - S_{n-1}$.
2. Calculate S_{16} by substituting $n = 16$ into the given formula.
3. Calculate S_{15} by substituting $n = 15$ into the given formula.
4. Subtract S_{15} from S_{16} to find a_{16} .

Derivation:

$$a_{16} = S_{16} - S_{15} = [3(16)^2 + 5(16)] - [3(15)^2 + 5(15)] = [3(256) + 80] - [3(225) + 75] = 848 - 750 = 98.$$

Arithmetic Progressions — General term from sum

- 23. 23.** The 4th term of an AP is 0. Prove that the 25th term of the AP is three times its 11th term.

Answer:

$$a_{25} = 3 \times a_{11}$$

Solution:

1. Express the 4th term as $a + 3d = 0$, leading to $a = -3d$.
2. Express the 25th term as $a + 24d$.

3. Express the 11th term as $a + 10d$.
4. Substitute $a = -3d$ into both expressions and compare.

Derivation:

$$\implies a = -3d. \quad a_{25} = a + 24d = -3d + 24d = 21d. \quad a_{11} = a + 10d = -3d + 10d = 7d. \quad \text{Since } 21d$$

Arithmetic Progressions — Properties of AP

- 24. 24.** How many terms of the AP 24, 21, 18, ... must be taken so that their sum is 78?

Answer:

4or13

Solution:

1. Identify $a = 24$, $d = -3$, $S_n = 78$. Use the quadratic sum formula $S_n = n/2[2a + (n - 1)d]$.
2. Substitute the known values to form a quadratic equation in n .
3. Solve the quadratic equation for n using factorization or the quadratic formula.
4. Verify the two values of n .

Derivation:

$$\begin{aligned} 78 &= \frac{n}{2}[2(24) + (n - 1)(-3)] \\ \implies 156 &= n(48 - 3n + 3) \\ \implies 156 &= 51n - 3n^2 \\ \implies 3n^2 - 51n + 156 &= 0 \\ \implies n^2 - 17n + 52 &= 0 \\ \implies (n - 4)(n - 13) &= 0. n = 4, 13. \end{aligned}$$

Arithmetic Progressions — Sum of n terms

- 25. 25.** Find the middle term of the Arithmetic Progression 7, 13, 19, ..., 241.

Answer:

124

Solution:

1. Identify $a = 7$, $d = 6$, $l = 241$. Find the total number of terms n using $l = a + (n - 1)d$.
2. Determine if n is odd or even to find the middle term position.
3. Calculate the value of the middle term(s) using the formula $a_k = a + (k - 1)d$.

Derivation:

$$\implies n = 40. \text{ Since } n = 40 \text{ is even, there are two middle terms: } 20^{th} \text{ and } 21^{st}. a_{20} = 7 + 19(6) = 121$$

Arithmetic Progressions — General term

- 26. 26.** The sum of the first n terms of an Arithmetic Progression is given by $S_n = 3n^2 + 5n$. Find the n^{th} term of this AP and hence find the 20^{th} term.

Answer:

$$6n + 2; 122$$

Solution:

1. Use the relation $a_n = S_n - S_{n-1}$ to find the general term.
2. Substitute $S_n = 3n^2 + 5n$.
3. Calculate $S_{n-1} = 3(n-1)^2 + 5(n-1)$.
4. Subtract S_{n-1} from S_n and simplify the expression.
5. Substitute $n = 20$ into the resulting formula to find a_{20} .

Derivation:

$$\begin{aligned} a_n &= S_n - S_{n-1} \\ a_n &= (3n^2 + 5n) - [3(n-1)^2 + 5(n-1)] \\ a_n &= 3n^2 + 5n - [3(n^2 - 2n + 1) + 5n - 5] \\ a_n &= 3n^2 + 5n - [3n^2 - 6n + 3 + 5n - 5] \\ a_n &= 3n^2 + 5n - 3n^2 + 6n - 3 - 5n + 5 = 6n + 2 \\ a_{20} &= 6(20) + 2 = 122 \end{aligned}$$

Arithmetic Progressions — Sum of n terms

- 27. 27.** In an AP, the sum of first 10 terms is -150 and the sum of its next 10 terms is -550 . Find the Arithmetic Progression.

Answer:

$$a = 4, d = -3; 4, 1, -2, -5, \dots$$

Solution:

1. Set up the equation for $S_{10} = -150$.
2. Recognize that the sum of the next 10 terms is $S_{20} - S_{10} = -550$.
3. Find $S_{20} = -550 + (-150) = -700$.
4. Solve the resulting system of two linear equations in a and d .
5. State the first few terms of the AP.

Derivation:

$$\begin{aligned} S_{10} &= 5(2a + 9d) = -150 \\ \implies 2a + 9d &= -30 \quad (1) \\ S_{20} &= 10(2a + 19d) = -700 \\ \implies 2a + 19d &= -70 \quad (2) \\ (2) - (1) : 10d &= -40 \\ \implies d &= -4 \quad (\text{Correction: } d = -4, \text{ then } 2a - 36 = -30, 2a = 6, a = 3) \\ \text{Re-calculating: } 2a + 9(-4) &= -30 \\ \implies 2a - 36 &= -30 \\ \implies 2a &= 6 \\ \implies a &= 3 \\ \text{AP: } 3, -1, -5, -9, \dots \end{aligned}$$

28. 28. A sum of Rs. 1600 is to be used to give 5 cash prizes to students of a school for their overall academic performance.

Answer:

Rs. 400, Rs. 380, Rs. 360, Rs. 340, Rs. 320

Solution:

1. Let the prizes be $a, a - 20, a - 40, a - 60, a - 80$.
2. Use the sum formula $S_n = \frac{n}{2}[2a + (n - 1)d]$.
3. Substitute $n = 5$, $S_5 = 1600$, and $d = -20$.
4. Solve for a .
5. List all 5 terms.

Derivation:

$$\begin{aligned}
 1600 &= \frac{5}{2}[2a + (5 - 1)(-20)] \\
 3200 &= 5[2a - 80] \\
 640 &= 2a - 80 \\
 2a &= 720 \\
 \implies a &= 360 \text{ (Wait, check: } 5a + 10d = 1600/2 \text{ is wrong)} \\
 1600 &= \frac{5}{2}(2a - 80) \\
 \implies 640 &= 2a - 80 \\
 \implies 2a &= 720 \\
 \implies a &= 360
 \end{aligned}$$

29. 29. Find the sum of all three-digit numbers which leave a remainder of 3 when divided by 5.

Answer:

98550

Solution:

1. Identify the first three-digit number a such that $a \equiv 3 \pmod{5}$, which is 103.
2. Identify the last three-digit number l such that $l \equiv 3 \pmod{5}$, which is 998.
3. Determine the number of terms n using $a_n = a + (n - 1)d$ where $d = 5$.
4. Calculate $n = \frac{l-a}{d} + 1$.
5. Calculate the sum using $S_n = \frac{n}{2}(a + l)$.

Derivation:

$$a = 103, l = 998, d = 5$$

$$998 = 103 + (n - 1)5$$

$$895 = (n - 1)5$$

$$n - 1 = 179$$

$$\implies n = 180$$

$$S_{180} = \frac{180}{2}(103 + 998) = 90(1101) = 99090$$

Arithmetic Progressions — Sum of n terms